Spring 2021 / EC571

Instructor:Ajay Joshi

Final-Exam

Date: May 7, 2021

Name : _____

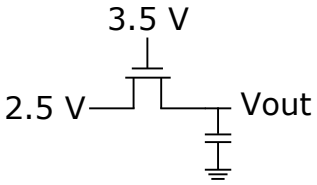
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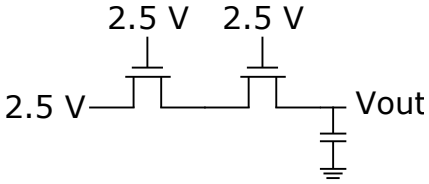
Question	Maximum points	Points scored
1	20	
2	20	
3	20	
4	20	
5	20	
6	20	
Bonus	5	
Total	100	

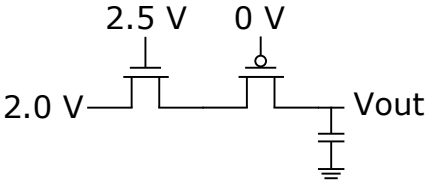
1. The duration of the exam is 2 hours.
2. The maximum grade is 100 points.
3. Please write your name on all pages.
4. Attempt any 5 of the 6 questions. Please indicate the question number that should not be graded.
5. If you need to use extra sheets of paper, write your name and BU ID on it, and clearly indicate the question number.
6. Material sharing is not allowed.
7. Calculators are allowed.
8. The exam is open book open notes.
9. Boston University Academic Code of Conduct applies.
10. BU ENG is collaborating with several online services, including Chegg, to monitor activity during exams and obtain account information and IP addresses connected to uploaded exam materials and their access. The penalties for cheating are extremely severe including expulsion from BU.

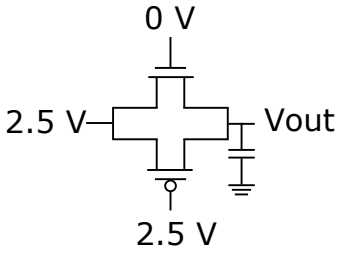
Question 1

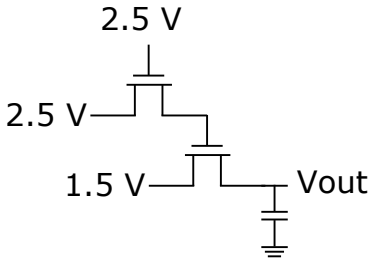
1. Find the final value of the voltage V_{out} for the following 5 configurations (2 points each). Assume $V_{th,N}$ = 0.5 V and $V_{th,P}$ = -0.5 V. Assume that the capacitor and any internal nodes are initially discharged. Ignore subthreshold conduction and body effect.









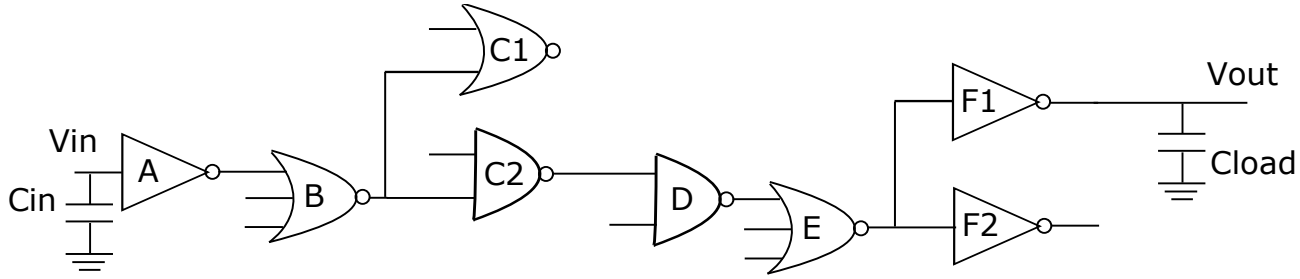


2. Draw the schematic for the static complementary CMOS implementation of the two functions listed below. For each design, state the size of the NMOS and PMOS transistors (as a function of W) such that the high-to-low delay and low-to-high delay of both designs will match the high-to-low delay and low-to-high delay, respectively, of the inverter. You can assume that the widths of the NMOS transistor and PMOS transistor of the inverter are W and 2W, respectively.

   (a) $F1(A,B,C) = \overline{A \cdot B + C}$ (5 points)

   (b) $F2(P,Q,R,S) = \overline{P \cdot Q + R \cdot S}$ (5 points)

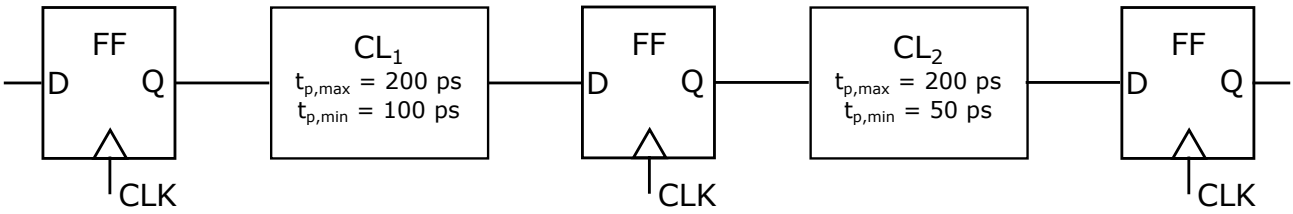
Question 2



For the circuit shown in the above figure, using the logical effort technique, calculate the input width of transistors for each gate in the path such that total delay of the path is minimized. You can assume gates F1 and F2 are identical. Note C1 and C2 are not identical. Assume $C_{in} = 10\ fF$, $C_G = 1\ fF/\mu m$, $C_{load} = 120\ fF$ and $2R_{sq(NMOS)} = R_{sq(PMOS)}$. (20 points)

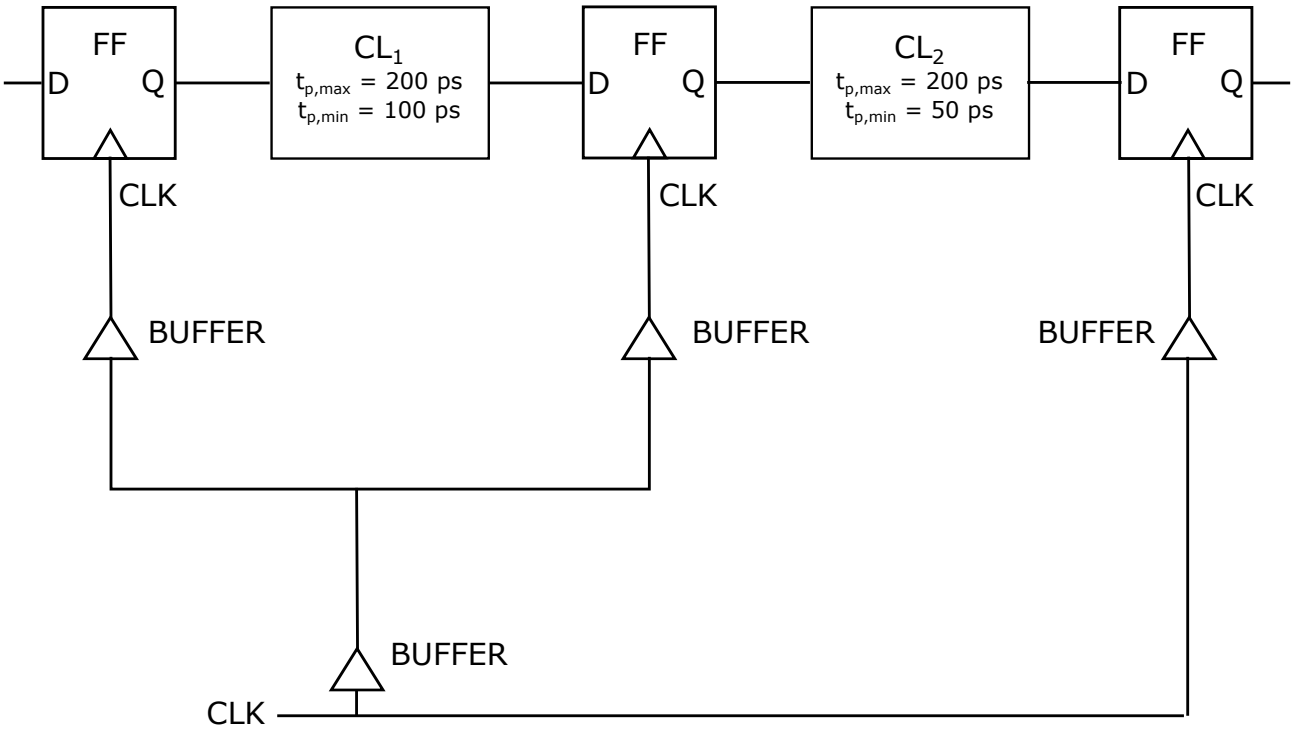
Question 3

Consider the pipelined sequential logic block shown below. The minimum and maximum propagation delay of each combinational logic block is annotated to each combinational logic block. Assume $t_{clk-to-q} = t_{setup} = 50\ ps$, and $t_{hold} = 75\ ps$.



1. What is the minimum clock period for the pipelined sequential logic block above? Assume there is no clock skew or jitter. Are there any hold-time violations? (7 points)

2. In the figure below we have added the clock distribution network. Assume each buffer has a delay that can vary by $\pm$ 10 ps. What is the minimum clock period in that case? Are there any hold-time violations? (8 points)

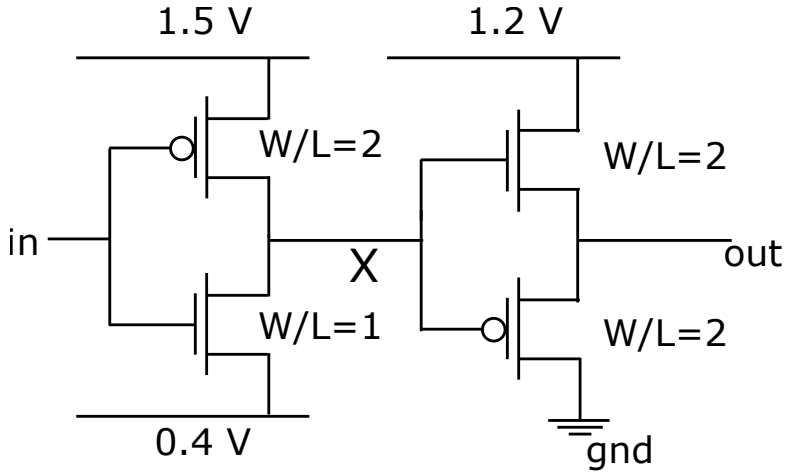


3. Explain the difference between a flip-flop and a latch. (5 points)

Question 4

1. An AI accelerator consumed 0.1 mW/MHz when fabricated using 65 *nm* process. The area of the accelerator is 0.3 mm^2. Assume a 1 GHz clock frequency and 1.0 V supply voltage, and we have short channel devices. What would be the power, area, frequency and power density of the accelerator if the accelerator design was scaled down to 45 *nm* with full scaling and general scaling (where supply voltage is scaled down to 0.8 V). Ignore leakage power scaling. (10 points)

2. Using the simple switch model of the transistors with $R_{sq,NMOS}$ = 13 Kohm/sq, $R_{sq,PMOS}$ = 26 Kohm/sq, $V_{th,N}$ = 0.25 V, $V_{th,P}$ = -0.25 V and V_{DD} = 1.5 V, draw the VTC of the circuit shown below. Assume V_{in} varies from 0 to 1.5V. Determine the values for V_{OH}, V_{OL}, V_{IH} and V_{IL}. Assume V_{out} = 0 V initially. Explain your answer in detail. (10 points)



Question 5

1. Compare (qualitatively) static complementary logic, pseudo-NMOS logic, pass-transistor logic and dynamic logic in terms of energy, performance, area, robustness and ease of design. (10 points)

2. Assume V_{dd} = 2.5 V, $V_{TH,p}$ = -0.5 V and no short-channel effects.

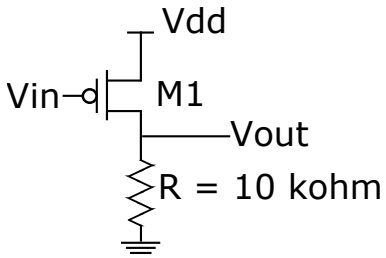


Figure 1

(a) For $V_{in} = V_{dd}$, what is the value of V_{out}? (1 point)

(b) For V_{in} = 0 V and V_{out} = 0 V, what is the region of operation for M1? (1 point)

(c) Write the equation for calculating V_{out} in that region (1 point)

(d) With V_{in} fixed at 0 V, V_{out} will start increasing. At what output voltage does the region of operation change for M1? (2 points)

(e) Write the equation for calculating V_{out} for this new region of operation (1 point)

(f) As V_{out} further increases, what happens to the current equation? (2 point)

(g) Write the equation for calculating V_{out} for this new region of operation (1 point)

(h) What happens if we increase or decrease the resistance value of the resistor in the circuit? (1 point)

Question 6

1. In this problem you have to design a minimum delay static decoder for an SRAM array with containing 128 rows and 16 columns (each wordline will drive 32 cells and each bitline has 128 cells on it). Each SRAM cell is a standard 6T cell with the two NMOS access transistors each having a width of 250 nm, the two NMOS transistors of the inverter each having a width of 250 nm, and the two PMOS transistors of the inverter each having a width of 500 nm. You can assume $C_G = C_D = 1\ fF/\mu m$, $R_{sq,NMOS} = 13\ Kohm/sq$, and $R_{sq,PMOS} = 26\ Kohm/sq$. Assume all transistors are long-channel transistors. Assume wire capacitance is negligible. Assume the input capacitance of each address line is $2\ fF$. Your final answer should show gate-level implementation. You do not need to determine the sizes of the individual transistors. (15 points)

2. Explain the concept read margin and write margin in SRAM cells. (5 points)

Bonus question

Name 5 cities in the southern hemisphere of earth.